Data Quality Report

Data validation, cleaning log and relationship integrity

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Scope and how to read this report

This report covers the Data Science track of the Week 6 brief for 20 facilities in the HealthConnect Rwanda network. It accompanies the executed notebook (HealthConnect_DataScience_Notebook.ipynb), which reproduces every table and figure here.

Three things shape every conclusion. First, five tables named in the brief were not supplied (Health Facilities master, Laboratory Tests, Prescriptions, Admissions, Referrals), so admission, length-of-stay, laboratory, prescription and referral analyses are replaced by labelled proxies or marked not computable. Second, only 5,000 of the 60,000 billing, diagnosis and payment records link to an appointment, so appointment-level work rests on 5,000 rows (1,756 for satisfaction). Third, the data behave like simulated data: variables are close to uniform and almost no relationship exists between them. Where a test finds nothing, this report says so.

Data inventory

27 CSV files were loaded (273,602 rows). The largest tables are Billing, Diagnoses and Payments (60,000 each), Insurance Claims (42,036), Patient Feedback (21,064), Patients (20,000) and Appointments (5,000). The two Departments files are identical copies.

Missing from the supply	Consequence
Health Facilities (master)	Facility IDs HF001-HF020 are inferred from Departments; no facility type, ownership, location or bed counts
Admissions	No admission, discharge or length-of-stay data; admission is modelled through a proxy
Laboratory Tests / Prescriptions	Laboratory and medication utilisation not computable
Referrals	Referral rate and destinations not computable

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The cleaning log records 42 findings: 6 Critical, 10 High, 12 Medium, 4 Low and 10 informational. Principle: flag, do not delete, unless a record is provably invalid. No rows were removed.

Validation checks required by the brief

Check	Result	Detail

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Foreign-key relationships	3 of 26 relationships have orphans	All orphans sit in the transaction chain (Billing, Diagnoses, Payments, Feedback -> Appointments); reference tables are clean
Duplicate Patient IDs	0	PatientID unique across 20,000 patients
Duplicate Appointment IDs	0	AppointmentID unique across 5,000 appointments
Invalid Facility IDs (outside HF001-HF020)	0	But facility on appointment disagrees with department's facility in 94.9% of rows
Invalid Department IDs	0	All 200 department IDs valid
Impossible ages	0	Ages 0-90; consistent with DOB. But 2594 children <12 have adult-range height/weight
Invalid billing amounts	0	All amounts positive; payments and claims never exceed bills; but Status labels contradict payments

Critical and high-severity findings

Table	Issue	Records	Action taken
appointments	Appointment facility differs from the facility that owns the appointment's department	4,744	Flagged. Facility-level analysis uses the recorded Appointment.FacilityID; department-level analysis uses department TYPE (10 types, identical across facilities). Ranking sensitivity tested in 5.3
billing	Bills whose AppointmentID is not in the Appointments table (orphans)	55,000	Retained in the 60k bill-level analysis; excluded from appointment-level modelling
diagnoses	Diagnoses whose AppointmentID is not in Appointments	55,000	Excluded from appointment-level analysis
payments	Payment rows whose bill's appointment is missing	55,000	As billing
feedback	Feedback rows whose AppointmentID is not in Appointments	19,308	Used only for hospital-wide rating distribution; facility/department satisfaction uses the linked rows
ALL	No Admissions, Length-of-Stay, Laboratory, Prescription or Referral tables	0	Admission modelled through a labelled proxy (4.4); LOS, lab, prescription and referral analyses marked not computable
patients	Phone saved in scientific notation (e.g. 2.50788E+11) -> only 6 significant digits survive; only 10 distinct values in 20,000 rows	20,000	Converted to text; Phone cannot identify duplicate patients or contact anyone - excluded from analysis
patients	Children under 12 with adult-range height/weight (height 140-190 cm, weight 40-120 kg recorded for all ages)	2,594	Height/weight/BMI treated as unreliable for under-18s: BMI_clean set to missing for Age<18
patients	Every patient (100%) has an InsuranceID (15 plans); there is no 'uninsured' category and no insurance master table	20,000	Insurance coverage analysed through claims filed at billing level instead (see 4.2)

Table	Issue	Records	Action taken
appointments	Appointments dated AFTER the analysis date (24 Oct 2026) yet carrying diagnoses, bills and feedback	874	Flagged (future_dated); retained because they are indistinguishable from the rest of the 2026 series - results shown with and without them where it matters
appointments	Doctor's home department differs from the appointment's department	4,968	Flagged; doctor workload computed per DoctorID regardless of department
billing	Bills marked 'Paid' whose payment is below the bill amount	4,042	Billing.Status treated as unreliable; Outstanding recomputed as Amount - AmountPaid
billing	Bills marked 'Pending'/'Partially Paid' that are fully paid	32,010	As above
feedback	Rating contradicts the text comment (e.g. rating 1 with 'Excellent', rating 5 with 'Poor')	6,336	Rating (numeric) used as the satisfaction measure; Comment ignored (Spearman rho Rating~Comment = 0.003)
revenue_summary	Revenue_Summary does not reconcile with Billing (summary total 1.13bn RWF vs 9.31bn billed across 60,000 bills; 60,809 summary appointments vs 5,000 appointment records)	12	Treated as a separate management series; both are modelled and compared
beds	All 2,000 beds have Status = 'Available' (no occupied beds recorded)	2,000	Bed occupancy/utilisation is NOT computable

Medium and low findings (phone numbers stored in scientific notation, every date of birth on 1 January, under-18s recorded as married or holding Master's degrees, missing AB- blood group, placeholder district names) are listed in outputs/data_quality_report.csv with the treatment applied.

Relationship integrity

Reference tables (geography, departments, rooms, beds, equipment, staff) are fully consistent. The problems sit in the transaction chain: 55,000 bills, diagnoses and payments point to appointment IDs that do not exist, and the facility, department and doctor recorded on an appointment agree with each other in only about 5% of rows (5.12% for facility vs department owner; 5.14% for facility vs doctor home facility). Facility-level results therefore use the recorded appointment facility, department-level results use department type, and the notebook tests ranking sensitivity.

Feature engineering

Required feature	Definition	Status
Patient Age Group	Banded Age (9 bands; 3 broad bands)	Done
High-Risk Patient Indicator	Age>=65 with a chronic condition, OR a chronic condition with any Severe diagnosis (rule-based, patient-level)	Done

Required feature	Definition	Status
Length of Stay Category	Requires admission/discharge dates (table not supplied)	NOT COMPUTABLE
Frequent Visitor Indicator	Patient with 2+ appointments in the data	Done
Insurance Coverage Category	Claim approved / Claim pending / Self-pay (no claim) - all patients hold an InsuranceID so claim presence is the usable signal	Done
Revenue Category	Tertiles of the 60,000 bill amounts (Low/Medium/High)	Done
Disease Severity Score	Mild=1, Moderate=2, Severe=3	Done
Facility Performance Score	50 + 10 x mean z-score of volume, mean rating, collection rate, claim approval rate	Done
Patient Satisfaction Category	Rating 1-2 Low, 3 Medium, 4-5 High (1,756 appointments with feedback)	Done
Department Workload Index	Appointments per rostered clinical staff, scaled so the network mean = 100	Done